

## GED Assignment 12 - Polynomials

1. Perform the required operations and simplify the following.

$$(a) \quad 7a^2 + 10y - 2xy + 16y - 4a^2 + 8xy$$

$$(b) \quad 14r^2x + 3rx^2 - 5rx - (6rx + 4r^2x - 2rx^2)$$

$$(c) \quad 3a(12a + 2ab - 6b)$$

$$(d) \quad (3x + 4)(5x - 2)$$

$$(e) \quad (x^2y - 2y)(2x - xy)$$

2. Fully factor the following.

$$(a) \quad 4x^2 + 6x - 10$$

$$(b) \quad 12a^2b^3 + 3ab^2 - 9a^2b + 15a^2b^2$$

$$(c) \quad x^2 - 6x + 8$$

$$(d) \quad t^2 + 8t + 16$$

$$(e) \quad 3x^2 + 14x - 5$$

$$(f) \quad 14p^2 + 17p - 6$$

$$\begin{aligned} 1. (a) \quad & 7a^2 - 4a^2 + 10y + 16y - 2xy + 8xy \\ & = 3a^2 + 26y + 6xy \end{aligned}$$

$$\begin{aligned} (b) \quad & 14r^2x - 4r^2x + 3rx^2 + 2rx^2 - 5rx - 6rx \\ & = 10r^2x + 5rx^2 - 11rx \end{aligned}$$

$$(c) \quad 36a^2 + 6ab - 12ab$$

$$\begin{aligned} (d) \quad & 15x^2 - 6x + 20x - 8 \\ & = 15x^2 + 14x - 8 \end{aligned}$$

$$(e) 2x^3y - x^3y^2 - 4xy + 2xy^2$$

$$+ (6)$$

$$\times (-40)$$

$$+10, -4$$

$$2. \quad (a) \quad 4x^2 + 6x - 10$$

$$= 4x^2 + 10x - 4x - 10$$

$$= 2x(2x+5) - 2(2x+5)$$

$$= (2x-2)(2x+5)$$

$$(b) \quad 12a^2b^3 + 3ab^2 - 9a^2b + 15a^2b^2$$

$$= 3ab(4ab^2 + b - 3a + 5ab)$$

$$(c) \quad x^2 - 6x + 8$$

$$+ (-6)$$

$$\times (8)$$

$$-4, -2$$

$$= (x-4)(x-2)$$

$$(d) \quad t^2 + 8t + 16$$

$$+ (8)$$

$$\times (16)$$

$$4, 4$$

$$= (t+4)(t+4)$$

$$= (t+4)^2$$

$$(e) \quad 3x^2 + 14x - 5$$

$$+ (14)$$

$$\times (-15)$$

$$15, -1$$

$$= 3x^2 + 15x - x - 5$$

$$= 3x(x+5) - (x+5)$$

$$= (x+5)(3x-1)$$

$$(f) \quad 14p^2 + 17p - 6$$

$$+ (17)$$

$$\times (-84)$$

$$21, -4$$

$$= 14p^2 + 21p - 4p - 6$$

$$= 7p(2p+3) - 2(2p+3)$$

$$= (2p+3)(7p-2)$$